

the case of theft," explains a paper on this technology.

Miniaturise

Sensor technology can be used to help miniaturise big laboratory equipment through advanced integration technologies.

Smaller labs. MICROMOLE project at Fraunhofer IIS is to reduce the size of a tabletop laboratory system down to an object that will fit in the hand, while being able to operate in wastewater without human intervention.

Main element of the system is the chemical sensor, which is miniaturised using micro-fluidic technology. It implements high sensitivity and selectivity, while being robust against variations in water properties due to a custom-designed sensitive electrode.

Pocket RADARs. One of the most exciting announcements made this year is Google's Soli Project. It

is a new sensing technology that uses miniature radar through an interaction sensor to detect touch-less gesture interactions. The sensor tracks sub-millimetre motion at high speeds with great accuracy.

The chip comes in an 8mm x 10mm package that emits electromagnetic waves in a broad beam and tracks the reflected beam for information. Soli sensor is a fully-integrated, low-power radar operating in 60GHz industrial, scientific and medical radio band. Infineon and Google will be jointly marketing this technology mid 2017 onwards.

Acti3D technology makes available algorithms for an optical sensor system that enables detection, localisation and analysis of movements without the output of real image data. Algorithms record important points and image features, independent of the ambient lighting. In a subsequent analysis, movements are classified and assigned to

the foreground or background. This way, people can be distinguished from technical devices.

The future of sensing

There are quite some exciting and award-winning developments happening in this space, too. Bosch's BMF055 won this year's Innovation Award at Sensors Expo & Conference as the industry's very first custom-programmable 9-axis motion sensor.

Another Innovation Award winner was AWAIBA's NanEye 2D sensor, which is a system on chip with complementary metal-oxide-semiconductor image sensors.

Third in line was Siargo's FI1000 time-of-flight MEMS flow sensor.

With the strong growth of the IoT, research and development in sensor technology is already seeing quite some hectic activity. Let us hope it will get even more exciting in the coming year. **EFY**



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